

# Decodelab Project 3

AI Recommendation Logic System

Developer: Philomina Paul-Ubah | Language: Python 3

## Project Key Concepts

Logic Building

Pattern Matching

Content Similarity

Recommendation System

## Complete Python Source Code

```
# Decodelab Project 3: AI Recommendation Logic System
# Developed by: Philomina Paul-Ubah

# 1. Dataset: Catalog of available items with associated tags/categories
CATALOG = [
    {"id": 1, "name": "Wireless Headphones", "category": "Electronics", "tags": ["tech", "music",
"audio", "gadgets"]},
    {"id": 2, "name": "Running Shoes", "category": "Footwear", "tags": ["sports", "fitness", "running",
"outdoors"]},
    {"id": 3, "name": "Python Programming Guide", "category": "Books", "tags": ["tech", "coding",
"education", "books"]},
    {"id": 4, "name": "Smart Fitness Watch", "category": "Electronics", "tags": ["tech", "fitness",
"gadgets", "health"]},
    {"id": 5, "name": "Yoga Mat", "category": "Fitness", "tags": ["fitness", "health", "sports",
"wellness"]},
    {"id": 6, "name": "Sci-Fi Novel", "category": "Books", "tags": ["books", "reading", "fiction",
"entertainment"]}
]

def get_user_preferences():
    """Captures user input for interests."""
    print("=====")
    print("      DECODELAB PROJECT 3: AI RECOMMENDATION      ")
    print("      Developer: Philomina Paul-Ubah      ")
    print("=====\\n")
    print("Available interests: tech, sports, fitness, books, gadgets, health, music, outdoors\\n")

    user_input = input("Enter your interests separated by commas (e.g., tech, fitness): ")
    interests = [interest.strip().lower() for interest in user_input.split(",") if interest.strip()]
    return interests

def calculate_similarity(user_interests, item_tags):
    """
    Matches preferences using logic/similarity scoring.
    Score = number of overlapping tags between user interests and item tags.
    """
    user_set = set(user_interests)
    item_set = set(item_tags)
    overlap = user_set.intersection(item_set)
    return len(overlap)

def recommend_items(user_interests, catalog, top_n=3):
    """Generates and displays recommended items based on similarity scores."""
    scored_items = []

    for item in catalog:
        score = calculate_similarity(user_interests, item["tags"])
        if score > 0: # Only consider items with at least one match
            scored_items.append((score, item))
```

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# Sort items by similarity score in descending order
scored_items.sort(key=lambda x: x[0], reverse=True)

return scored_items[:top_n]

def display_recommendations(scored_items):
    """Displays the recommendations to the user."""
    print("\n--- Decodelab Recommended Items ---")
    if not scored_items:
        print("No matches found based on your interests. Try different keywords!")
        return

    for rank, (score, item) in enumerate(scored_items, start=1):
        print(f"{rank}. {item['name']} | Category: {item['category']} (Match Score: {score})")

# Run the project workflow
if __name__ == "__main__":
    user_interests = get_user_preferences()
    if user_interests:
        recommendations = recommend_items(user_interests, CATALOG)
        display_recommendations(recommendations)
    else:
        print("No interests provided. Exiting Decodelab Project 3.")

```

## How to Execute

- Save the source code above to a file named `decodelab_project_3.py`.
- Run in terminal using `python decodelab_project_3.py`.
- Input your preferred topics separated by commas to get recommendations.